

Chapter

Measure Of Central Tendency

1. The median of the following observations arranged in ascending order is 64.
27, 31, 46, 52, x , $x + 4$, 71, 79, 85 90 , Find the value of x : Main 2024

- (a) 60
- (b) 61
- (c) 62
- (d) 66

2. For the given 25 variables: $x_1, x_2, x_3, \dots, x_{25}$

Assertion (A): To find median of the given data, the variate needs to be arranged in ascending or descending order.

Reason (R): The median is the central most term of the arranged data.

- (a) A is true, R is false
- (b) A is false, R is true
- (c) both A and R are true
- (d) both A and R are false

SQP 2024

3. Which of the following cannot be determined graphically for a grouped frequency distribution? Main 2023

- (a) Median
- (b) Mode
- (c) Quartiles
- (d) Mean

4. The following table gives the marks scored by a set of students in an examination. Calculate the mean of the distribution by using the short cut method. Main 2023

Marks	Number of workers (f)
0–10	3
10–20	8

20–30	14
30–40	9
40–50	4
50–60	2

5. Find the mean of the following distribution by step deviation method :
Comp 2023

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	5	13	20	15	7	5

6. The mean of the following distribution is 50. Find the unknown frequency
Main 2022

Class Interval	0-20	20-40	40-60	60-80	80-100
Frequency	6	f	8	12	8

7. The mean of the following distribution is 314. Determine the missing frequency x .
Comp 2022

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	5	x	10	12	7	8

Solution

1.

Here, number of observations, $n = 10$, which is even.

$$\text{Median} = \frac{\left(\frac{n}{2}\right)\text{th observation} + \left(\frac{n}{2} + 1\right)\text{th observation}}{2}$$

$$64 = \frac{\left(\frac{10}{2}\right)\text{th observation} + \left(\frac{10}{2} + 1\right)\text{th observation}}{2}$$

$$64 = \frac{5\text{th observation} + 6\text{th observation}}{2}$$

$$64 = \frac{x + x + 4}{2}$$

$$64 = \frac{2x + 4}{2}$$

$$64 = x + 2$$

$$60 = x$$

Thus (a) is correct option.

2. (c) both A and R are true

3. (d) Mean

- **Mean** → **cannot** be determined graphically

4.

Marks	Number of Students f_i	Class marks x_i	$d_i = x_i - A$	$f_i d_i$
0–10	3	5	–20	–60
10–20	8	15	–10	–80
20–30	14	<u>25</u>	0	0
30–40	9	35	10	90
40–50	4	45	20	80
50–60	2	55	30	60
	$\Sigma f_i = 40$			$\Sigma f_i d_i = 90$

$$\begin{aligned}
 \text{Mean} &= A + \frac{\sum fd}{\sum f} \\
 &= 25 + \frac{90}{40} \\
 &= 25 + 2.25 \\
 &= 27.25
 \end{aligned}$$

✓ Mean of the distribution = 27.25 marks

5.

Class	x_i	$u_i = \frac{x_i - a}{h}$	f_i	$f_i u_i$
0-10	5	-3	5	-15
10-20	15	-2	13	-26
20-30	25	-1	20	-20
30-40	35	0	15	0
40-50	45	1	7	7
50-60	55	2	5	10
			$\Sigma f_i = 65$	$\Sigma f_i u_i = -44$

Let assumed mean, $a = 35$ and given $h = 10$.

$$\begin{aligned}
 \text{Mean, } M &= a + \frac{\sum f_i u_i}{\sum f_i} \times h \\
 &= 35 + \frac{-44}{65} \times 10 \\
 &= 35 - 6.76 = 28.24
 \end{aligned}$$

6.

Class marks: 10, 30, 50, 70, 90

Mean = 50

$$50 = \frac{6(10) + f(30) + 8(50) + 12(70) + 8(90)}{6 + f + 8 + 12 + 8}$$

$$50 = \frac{2020 + 30f}{34 + f}$$

$$1700 + 50f = 2020 + 30f \Rightarrow 20f = 320 \Rightarrow f = 16$$

$$\boxed{f = 16}$$

7.

Class intervals: 0–10, 10–20, 20–30, 30–40, 40–50, 50–60

Frequencies: 5, x , 10, 12, 7, 8

Class marks:

5, 15, 25, 35, 45, 55

Mean = 31.4

$$31.4 = \frac{5(5) + x(15) + 10(25) + 12(35) + 7(45) + 8(55)}{5 + x + 10 + 12 + 7 + 8}$$

$$31.4 = \frac{1450 + 15x}{42 + x}$$

$$31.4(42 + x) = 1450 + 15x$$

$$1318.8 + 31.4x = 1450 + 15x$$

$$16.4x = 131.2 \Rightarrow x = 8$$

$$\boxed{x = 8}$$